Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering *CGPA: 9.05*

Projects

Negotiate, Then Solve: LLM Agents for Timetabling | *Python, FastAPI, React, TypeScript, Gemini, RAG*

- Built an LLM assistant that turns scheduling requests into constraints for a solver and negotiates conflicts with people.
- Shipped it as a React + FastAPI app with role-based access and an approval queue, tested with **215 pytest tests**.
- Added search over policy documents and safety checks; **no attack succeeded** in 50 prompt-injection tests.
- Fine-tuned a small 4B model with LoRA that outperformed the Gemini model it learned from.

IndicLLM: Transformer Pretraining & Reasoning SFT | *Python, PyTorch, ONNX, FastAPI, Docker*

- Pretrained GPT-style language models for Hindi and Assamese from scratch in PyTorch, including the data and tokenizers.
- Fine-tuned them for multi-step reasoning, reaching **96%** test accuracy against a 48% baseline.
- Exported the models to run fully in the browser and deployed them as a live demo with Docker and FastAPI.

Peer-to-Peer Distributed File Sharing System | *C++, Multithreading, Sockets, OpenSSL, Linux*

- Built a BitTorrent-style file sharing system in C++ with a central tracker, multithreaded peers and per-piece integrity checks.
- Profiled and fixed bottlenecks in scheduling and networking, cutting a 1GB download from **193 s to under 13 s**.

Warehouse Traffic Control: Distributed Robot Coordination | *JavaScript, Three.js, Web Workers, Vite*

- Built a live 3D browser simulator of a 200-robot warehouse in Three.js, deployed to GitHub Pages.
- Designed a lease-based protocol with **zero collisions** over 416K moves, even with lost messages, clock drift and crashes.
- Added congestion-aware routing that nearly doubled the tasks completed at 150 robots.

Technical Skills

Languages: Python, JavaScript, TypeScript, C/C++, SQL, HTML/CSS.
Web & Backend: React, Next.js, Node.js, FastAPI, PostgreSQL, SQLite, Three.js.
AI: LLM APIs (Gemini), RAG, LoRA, PyTorch.
Tools: Git, GitHub Actions, Docker, Vercel, Railway, pytest, Linux.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd by GPA in the B.Tech CSE batch at IIIT Ranchi.
- **PGEE 2025:** Secured All-India Rank 100 in the IIIT Hyderabad Post Graduate Entrance Examination.
- **Megathon 2025:** Placed 3rd in the mobile computing track.

Interests

- Reading web novels and manga.
- Playing chess and badminton.